Northstar Operations Room

WYRESTORM | TECHNICAL PROPOSAL

Document control

Reference
WYR-WWM800
Status
Draft proposal — subject to design and commercial approval
Prepared by
Solutions Team
Issue date
11 September 2026
Revision
2026-09-11T00:00:00.000Z
Product data
Product data f6cb16fe19ce · assembled 2026-10-30 · build unversioned

Executive Summary

Route two operator sources to four displays with resilient 4K distribution.

This document explains what we recommend, how it will work day to day, the equipment involved, and what still needs confirming before it becomes a final proposal.

Room at a glance

Operations room

Route two operator sources to four displays with resilient 4K distribution.

Room and design details

Room dimensions, capacity and site conditions require confirmation.

What this space delivers

Operational continuity: Operators retain controlled access to approved sources.

In this room

Items are grouped by the job each one does in the room, and the number matches the equipment schedule further on. Anything marked optional is still your decision, and work by others is listed under Complete System Design and Responsibilities.

Encode operator sources

ITEM	MODEL	DESCRIPTION	QTY	STATUS
1	NHD-600-E-TX	NetworkHD 600 encoder	2	Required

Decode display feeds

ITEM	MODEL	DESCRIPTION	QTY	STATUS
2	NHD-600-E-RX	NetworkHD 600 decoder	4	Required

Routing control

ITEM	MODEL	DESCRIPTION	QTY	STATUS
3	NHD-CTL-PRO-V2	NetworkHD controller	1	Required

Solution Overview

NetworkHD

What successful delivery looks like

Every approved source-to-display route passes functional testing.

What You Asked For

Route two operator sources to four displays with resilient 4K distribution.

Discovery Conversation

Below is the conversation this proposal is based on: the question we asked, the answer we recorded, and your own words where we captured them. "Confirmed with customer" means we checked that point with you; "To be confirmed" means it is still open and will be settled before the design is signed off.

QUESTION ASKED	ANSWER WE RECORDED	CUSTOMER WORDING	STATUS	CAPTURE CONFIDENCE
How many displays must receive independently routed content?	Four displays	Each operator needs independent routing.	Confirmed with customer	High confidence (12)

"Capture confidence" shows how closely your own words matched the answer we recorded. High confidence is a close match. Matched is a close match to a standard phrase. Low confidence — verify before quote is a partial match we will check with you before quoting. The number in brackets is the match score.

If a row has a note but no clear answer, that point was still open when this was written and will be confirmed before the design is signed off.

Recommended Solution

Recommendation: NetworkHD is the best fit for this requirement on the evidence gathered so far. Confidence: 96 out of 100.

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

Still to confirm: redundant power and switch failover behaviour during commissioning.

Required WyreStorm products: NHD-600-E-TX - NetworkHD 600 encoder; NHD-600-E-RX - NetworkHD 600 decoder; NHD-CTL-PRO-V2 - NetworkHD controller

Practical Operation

This is what the system will do for the people using the room, day to day.

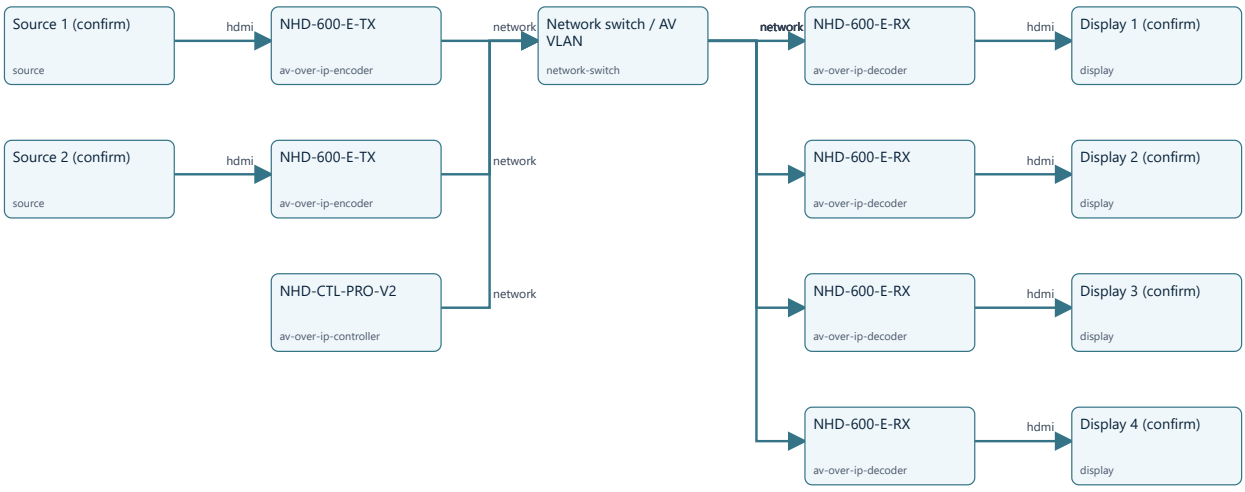
Operators select an approved source and route it to any display.

Commercial and Operational Benefits

A consistent, repeatable connection experience across the room(s) covered by this proposal.
AV-over-IP provides a potential expansion path, subject to compatible endpoints, network capacity and the final design.
We will tie these benefits to your own business measures before this proposal is issued.

Technical Architecture

The diagram and cable schedule below show how the system connects together. Both are built from the equipment schedule further on.



Concept drawing: confirm physical connections and site interfaces before installation.

Transport is shown generically because signal distance or connector evidence is not confirmed. Connections without a verified interface profile are proposed topology only. Confirm named ports, endpoint pairing, power and cable types before installation.

Control routing requires design: Specify the controller and controlled-device interfaces before allocating control cables.
Cable length not confirmed for hdmi run: The local source to encoder connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
Cable length not confirmed for hdmi run: The local source to encoder connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
Cable length not confirmed for hdmi run: The Decoder to display connection uses hdmi transport (max 15m). Confirm the actual cable run length is within spec before quoting.
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Power source unconfirmed for NHD-600-E-TX: "NetworkHD 600 encoder" (NHD-600-E-TX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
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Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-600-E-RX: "NetworkHD 600 decoder" (NHD-600-E-RX) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.
Power source unconfirmed for NHD-CTL-PRO-V2: "NetworkHD controller" (NHD-CTL-PRO-V2) has no power connection in the schematic. Confirm whether it is PoE-powered, has an included PSU, or requires a separate power supply.

This diagram shows the signal path we expect, from source to screen, so you and the installer are looking at the same picture. The cable schedule below gives the cable type, connectors, maximum length and what still needs checking for each connection.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
Source 1 (confirm) → NHD-600-E-TX	Video	15m	SOURCE PORT TBD to HDMI IN	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
Source 2 (confirm) → NHD-600-E-TX	Video	15m	SOURCE PORT TBD to HDMI IN	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-E-TX → Network switch	Network / AVoIP	Not confirmed	AV LAN (confirm) to SWITCH PORT 1	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-E-TX → Network switch	Network / AVoIP	Not confirmed	AV LAN (confirm) to SWITCH PORT 2	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-E-RX	Network / AVoIP	Not confirmed	SWITCH PORT 3 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
Network switch → NHD-600-E-RX	Network / AVoIP	Not confirmed	SWITCH PORT 4 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-E-RX	Network / AVoIP	Not confirmed	SWITCH PORT 5 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
Network switch → NHD-600-E-RX	Network / AVoIP	Not confirmed	SWITCH PORT 6 to AV LAN (confirm)	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-CTL-PRO-V2 → Network switch	Network / AVoIP	Not confirmed	AV LAN (confirm) to SWITCH PORT 7	Needs site confirmation	Confirm switch port availability, VLAN assignment, PoE budget, cable category, and network topology.
NHD-600-E-RX → Display 1 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.

CABLE RUN	TRANSPORT	MAX LENGTH	CONNECTORS	STATUS	VALIDATION REMINDER
NHD-600-E-RX → Display 2 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-E-RX → Display 3 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
NHD-600-E-RX → Display 4 (confirm)	Video	15m	HDMI OUT to DISPLAY INPUT (confirm)	Needs site confirmation	Confirm source and display connector types, cable length (typically <5m passive), HDCP version, EDID behaviour, and resolution/HDR requirements.
AV network infrastructure	Network / AVoIP	Not confirmed	RJ45 to RJ45	Needs site confirmation	Confirm switch model, port count, PoE budget, VLAN configuration, multicast policy, and uplink capacity.

Products in the Solution

Product photograph unavailable.

NHD-600-E-TX — NetworkHD 600 encoder
Role in this design: Encode operator sources

Official product link to be confirmed.

Product photograph unavailable.

NHD-600-E-RX — NetworkHD 600 decoder
Role in this design: Decode display feeds

Official product link to be confirmed.

Product photograph unavailable.

NHD-CTL-PRO-V2 — NetworkHD controller
Role in this design: Routing control

Official product link to be confirmed.

Complete System Design and Responsibilities

The items below complete the wider solution. Each one is still open: none of it is equipment or work included in this proposal, and we will agree the owner, detail and commercial treatment with you before it is issued.

Displays | Operator displays — Confirm size and mounting. | Qty 4 | Responsibility: Integrator | Scope status: allowance

Display Schedule

4 displays in scope, each driven by one feed.

SCREEN	RESOLUTION	FEEDS	SCALING	NOTES
Display 1	To confirm	1	Native 1:1, confirm at survey	Operator displays
Display 2	To confirm	1	Native 1:1, confirm at survey	
Display 3	To confirm	1	Native 1:1, confirm at survey	
Display 4	To confirm	1	Native 1:1, confirm at survey	

Basis: Native resolution, feed count and scaling are confirmed when the display models are recorded.

Equipment Schedule

ITEM	SKU	DESCRIPTION	ROLE	QTY	TYPE	NOTES
1	NHD-600-E-TX	NetworkHD 600 encoder	Encode operator sources	2	Required	Install in the equipment rack.
2	NHD-600-E-RX	NetworkHD 600 decoder	Decode display feeds	4	Required	One decoder per display.
3	NHD-CTL-PRO-V2	NetworkHD controller	Routing control	1	Required	Confirm firmware compatibility.
EQUIPMENT SCHEDULE TOTALS				7	3 line items	

Evidence basis: NHD-600-E-TX: Two confirmed operator sources | NHD-600-E-RX: Four independently routed displays | NHD-CTL-PRO-V2: Central routing and control requirement

Power Strategy

The power figures below come from WyreStorm product data for each selected product. They are the maximum the published specification states, and should be checked against the current datasheet before ordering.

SKU	QTY	PER-UNIT MAX	TOTAL MAX	POWER DATA (WYRESTORM)
NHD-600-E-TX	2	Not proven	—	802.3at PoE+
NHD-600-E-RX	4	Not proven	—	802.3at PoE+
NHD-CTL-PRO-V2	1	15.4W	15.4W	12V DC | PoE compliant; Max 15.4W, average 8W

Stated maximum consumption: approximately 15W across 1 product with proven figures. 3 products (NHD-600-E-TX, NHD-600-E-RX, NHD-CTL-PRO-V2) require PoE/PoH — confirm the injector or switch PoE budget covers the total.

Options

No optional enhancements are currently selected.

Assumptions and Exclusions

Assumptions

Assumption: Customer IT will provide a dedicated 10GbE AV VLAN.

Exclusions

Display, projector, LED and mounting hardware supply unless listed as a specific line item above.
Electrical, structural and builder's work.
Network switch procurement, VLAN/multicast configuration and IT commissioning access.
Structured cabling installation and certification.
Third-party control programming beyond WyreStorm-native presets.
Installation, labour and commissioning, unless separately quoted.

Risks and Dependencies

Risks and open questions

Risk: Confirm redundant power and switch failover behaviour during commissioning.

Dependencies

Dependency: NET-SWITCH-10G-MANAGED - Managed 10GbE network switch: Confirm switch model, port budget, VLAN and multicast configuration. Why: NetworkHD endpoints require managed 10GbE switching.

Implementation and Next Steps

Approve the network design and complete acceptance testing.

> This draft includes the selected WyreStorm products. We will confirm final specifications, accessories and dependencies before it is issued.

> Competitor products are not included in this proposal or its equipment schedule.

Important Notice & Best-Efforts Disclaimer

This proposal has been prepared on a best-efforts basis using WyreStorm Wingman to support your sales and design process. The information contained herein is provided for planning purposes only and does not constitute a binding quotation, order confirmation, or statement of capability.

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Technical Validation: System designs, power budgets, cable schedules, and equipment selections shown in this proposal are based on available product data and assumptions stated within. Final designs must be validated by a qualified AV integrator or pre-sales engineer before installation.

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Confirmed Requirement

Route two operator sources to four displays with resilient 4K distribution.

Design Assumptions

Customer IT will provide a dedicated 10GbE AV VLAN.

Recommended Architecture

NetworkHD is our recommended product family.

The confirmed routing and resilience requirement needs a scalable 10GbE fabric.

Required WyreStorm Products

NHD-600-E-TX - NetworkHD 600 encoder
NHD-600-E-RX - NetworkHD 600 decoder
NHD-CTL-PRO-V2 - NetworkHD controller

Optional Enhancements

No optional enhancements are currently selected.

Risks / Needs Validation

Confirm redundant power and switch failover behaviour during commissioning.

Next Steps

Approve the network design and complete acceptance testing.